

UNITEDHEALTH GROUP

Sample Invention Disclosure Form

Supplemental Document

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This form can be used to supplement the online [Invention Submission & Management](#) (ISM) form.
If using this form attach it to the online IDF draft before submission.



Instructions

- Provide as much detail as possible. Attach additional explanatory documents or materials (e.g. technical specifications, drawings, flowcharts, manuals, PPT presentations).
- Timing is critical to patent protection.
 - Talk to IP Legal Before Making a Public Disclosure.*** - If the invention is patentable, premature publication, commercial activity or other non-confidential disclosure may damage or preclude patent protection for your invention in many countries, including the United States.
 - Disclose As Soon As The Invention Is Clearly Conceptualized.*** - You do not need to wait until the invention is completely developed to seek patent protection. Under the "First to File" patent system in the U.S., filing as early as possible is preferable, even critical, to obtaining effective patent protection. This is especially true where others are filing patent applications in the same or adjacent technology areas.
- Questions? Contact IPLegal (iplegal@uhg.com).

Invention Description

- 1.1 Describe (a) the problem to be solved or addressed by the invention and (b) any known solutions and drawbacks to those existing solutions. Please cite any relevant technical documents. For software related inventions, describe whether there is a technical problem (e.g., in the field of computers or technological arts) that is being solved or addressed by the invention.

Background

UHG receives more than 100,000 member calls every day when considering all of its various lines of business. To determine call quality (which influences NPS scores), payors conduct a *post-facto* review that involves human agents listening to the calls. One aspect of this audit involves identifying problems. For example, if the member was upset, was the source of the problem something that occurred before initiating the call? Or did the agent say something during the call that upset the member? The results of this problem analysis are used for agent training and provide insights that improve the NPS. It is impossible to manually audit every call recording. Instead, the audit team relies on automated cognitive tools to identify the calls that would most likely affect UHG's NPS and then select some of these calls for further manual audit.

(a) Problem to be solved

Conventional machine learning-based tools assess the sentiment of a transcript by first segmenting a transcript into multiple text blocks based on a delimiter (e.g., a period), and then applying a sentiment ML model to identify the sentiment of each of these multiple text blocks. The numerical representation of each sentiment is simply averaged to arrive at a final score that reflects the overall sentiment of the entire transcript. However, in comparison to actual sentiments or "ground truths" (which are manually determined by a qualified person who audits a call recording), the overall sentiment is often inaccurate. For example, consider a real-life phone conversation below:

1. Agent: Good morning, thank you for calling our company
2. Caller: I have had multiple conversations with your support.
3. Agent: May I know your claim id?
4. Caller: It has been a complete waste of time.
5. Caller: No one has been able to address my problem.

Conventional tools generally split the transcript above in three simple static text blocks on which sentiment analysis is performed individually at each of the text blocks.

Simple-Block-1: I have had multiple conversations with your support.

Analyzed Sentiment: Positive (numerically represented as 1)

Simple-Block-2: It has been a complete waste of time.

Analyzed Sentiment: Neutral (numerically represented as 0)

Simple-Block-3: No one has been able to address my problem.

Analyzed Sentiment: Negative (numerically represented as -1)

A. Section Guide

Reviewers of this document will be looking for background on the invention to establish the context for the next sections.

B. Section Guide

*Reviewers of this document will be looking for a **technical problem** (i.e., **computer-related problem**) that you are solving with this invention. Tell us why the technical issue exists and why it needs to be fixed.*

*Note that a **technical problem** is not the same as a **business problem**. For example, an example business problem for this solution is that UHG is losing potential revenue in not being able to assess the sentiment of a member's call.*

*In contrast, an example **technical problem** for this solution is that conventional sentiment computer techniques analyze static text blocks in a mutually exclusive manner, and thus a computer cannot capture sentiment progression that spans across multiple and adjacent text blocks.*

Other examples of technical problems (which may or may not apply to this particular IDF) include:

- *Known computer methods are slow due to large amounts of data being processed;*

However, the simple average score of 0 would indicate an inaccurate sentiment of “neutral.” Accordingly, conventional tools perform sentiment analysis on text blocks in a mutually exclusive manner (intra-block), and thus are unable to capture real sentiment progression (inter-blocks). The inaccuracies in the simple average score impact the selection of call recordings for the manual audit process. A bad call selection further influences the quality of the audit. The disclosed method solves this problem.

In summary, this invention proposes methodology to solve the following technical problems in the realm of machine learning:

1. There is a large variation in the sentiment scores reported by various existing tools. The source of the high inaccuracy is due to lack of consideration and incorporation of a broader context of the conversation as a factor for analysis by the sentiment ML model.
2. Difficulty or inability for the sentiment ML model to capture sentiment progression across multiple text blocks.

(b) Known solutions

There are existing sentiment analysis solutions, such as <insert any product if known, e.g., Microsoft Azure, insert any hyperlink to a published paper, if known>. However, none of the existing solutions address the problems mentioned above because all these solutions perform sentiment analysis at each of the statically defined text blocks in a mutually exclusive manner. Section 1.3 further elaborates on the differences between existing solutions and our solution.

- Existing computer components required to carry out the process were previously not interoperable;
- Current computer processes utilize large amounts of computer memory; etc.

Tip: provide an example that clearly shows the technical problem. In this case, an example transcript between an agent and a caller helps inform reviewers about the limitations of conventional tools that generally split the transcript into static text blocks.

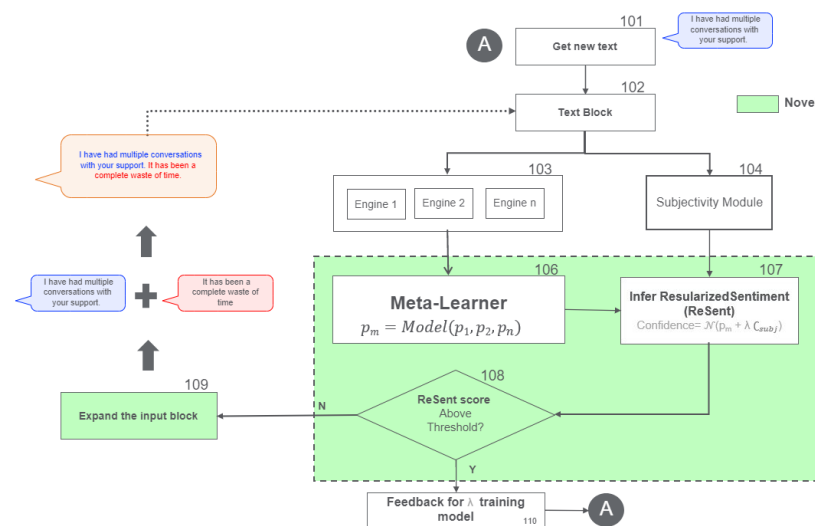
C. Section Guide

Reviewers of this document will be looking for a description of known solutions if you are aware of any (e.g., existing commercial offerings, published papers, etc.) and how they fall short of solving the technical problem. However, please do not identify a patent reference.

- 1.2 Describe the invention, including how it solves the problem described in Section 1.1. Describe how the invention works, in a manner that someone having the requisite technical experience could implement or practice the invention. For software-related inventions, describe how the invention provides a technical solution to a technical problem (e.g., in the field of computers or technological arts). If possible, include all essential structural/system components and process steps. Include or attach any relevant flowcharts, drawings, technical specifications, presentations, and other supporting documents.**

Detailed Description

(a) Illustration of overall schematic (new aspects indicated in green)



D. Section Guide

Reviewers of this document will be looking for **specific implementation details**, which should include the new aspect(s) (i.e., something that hasn't been done before).

Tip: Share visuals (e.g., schematic, graphs, or other documents). Visual representations of ideas get your message across easier and quicker than words alone.

Tip: Clearly indicate what is new in the visual. In this case, the green boundaries indicate the new aspects.

(b) Definitions

Sentiment: Sentiment has two aspects: class and confidence. As reported by known tools, sentiment has three classes: positive, neutral, negative. Each class of sentiment is accompanied by a confidence score ranging from 0 to 1. The respective confidence scores for the positive, neutral, and negative sentiments add to 1. The prevailing sentiment is the one associated with the greatest confidence score. For example, if sentiment analysis scores for positive, neutral, and negative sentiments are 0.9, 0.07, and 0.03, respectively, the prevailing sentiment is positive.

Elastic text block (new): This is a dynamically resizable text block, which could either be expanded by appending a new input text utterance or contracted by removing some existing text. In most cases, the elastic text block will expand relative to its original size.

Speech-to-text (STT): Speech to text conversion tools take audio as an input and return text for that input using speech recognition technology.

Subjectivity: This describes the subjective feelings, taste, or opinion of the input text being analyzed. A higher subjectivity indicates higher intensity of a person's opinion in a text, which corresponds to a stronger sentiment (e.g., a more positive sentiment, a more negative sentiment). A subjectivity engine, typically a classification neural network, outputs a score ranging between 0 to 1, as explained in the steps below.

Regularized Sentiment Score (new): This is a composite score that presents a sophisticated measure that considers sentiment and subjectivity.

(c) Method steps

1. Conventional STT engines convert the spoken audio recording of a call into text [steps 101-102 in the figure above].
2. The conventional STT engines transmit the first text block (e.g., “I **have had multiple conversations with your support.**”) to multiple known sentiment analysis engines of different types [step 103].
3. Each of the sentiment analysis engines independently infers and determines the sentiment of the first text block with a certain probability.
4. A meta-learning model receives each probability output of the sentiment analysis engines, learns from these outputs, and generates a localized sentiment probability (p_m) for the first text block [step 106]. For example, the localized sentiment probability (p_m) is (pos, neu, neg): (0.333, 0.333, 0.333). This way, the meta-learning model solves the first technical problem identified in section 1.1.
5. In parallel to step 2, the conventional STT engines also transmit the first text block to the subjectivity engine [step 104].
6. The subjectivity engine (e.g., a conventional classification neural network) determines the localized subjectivity of the sentiment of the text block with a certain probability (C_{subj}). For example, the localized subjectivity probability (C_{subj}) for the first text block is close to 0.
7. A regularized sentiment scoring module receives both (a) the localized sentiment probability (p_m) for the first text block output by the meta-learning model described in step 4 and (b) the localized subjectivity probability (C_{subj}) for the first text block described in step 6 to generate a quantifiable measure called Regularized Sentiment Score (RSS) [step 107]. The RSS is calculated as $N(p_m + \lambda C_{subj})$,

E. Section Guide

This section is optional, but it may be very helpful for reviewers' understanding to define important words used throughout the description. This is true especially for inventions in technical fields that are rapidly evolving (e.g., AI/ML). Always assume that reviewers are not familiar/experts within the technical field of your invention. Point out which words may be new (e.g., “elastic text block”). The new words most likely align with the new aspect(s) of your invention.

F. Section Guide

Provide technical details within the context of the methodology.

- *Provide enough detail so that reviewers will understand whether the idea is workable;*
- *Elaborate on which step(s) provide unique advantages, and why the step(s) solve the technical problems that you have identified;*
- *Identify exactly the components of the invention that you believe are new and set the invention apart from other solutions in the field.*

*Tip: Avoid describing only the result of the invention, and instead, describe **how** the result is achieved. For example, writing “**We** capture the sentiment and subjectivity of a text block and expand it” does not inform reviewers about **which component(s)** in the overall schematic capture sentiment/subjectivity and expand the text block, **how** the sentiment and subjectivity is captured, or **how** the text block is expanded.*

Tip: Provide step numbers in the method steps, and cross-reference

where λ is a penalizing factor ($\lambda \in [0,1]$) to compensate for high sentiment probability even when subjectivity is low, and N is a normalization constant.

8. The regularized sentiment scoring module checks if the RSS is above a certain predetermined threshold [step 108]. If the RSS is above the threshold, the regularized sentiment scoring module outputs p_m as the overall sentiment. If the RSS does not meet the threshold, the current text block is unable to represent the overall sentiment. This is where the method expands the text block (i.e., elastic text block) and evaluates the elastic text block. For example, the next text block from the transcript (e.g., “It has been a complete waste of time.”) is added to text block #1, so that the elastic text block becomes “I have had multiple conversations with your support. It has been a complete waste of time.” The process repeats starting at step 2. Accordingly, the change in size of the elastic text block depends on whether the RSS is above the threshold. This way, the regularized sentiment scoring module solves the second technical problem identified in section 1.1.

Example of improvement over existing tools

With reference to the example transcript shown above in Section 1.1, the machine learning model dynamically expands the initially defined text block (“Simple-Block-1”) to encompass more text, namely text in “Simple-Block-2” and “Simple-Block-3”:

Elastic-Block-1: I have had multiple conversations with your support.

Subjectivity ~ 0 – **threshold not met, continue resizing block**

New Sentiment (pos, neu, neg): (0.333, 0.333, 0.333)

Elastic-Block-2: I have had multiple conversations with your support. It has been a complete waste of time.

Subjectivity =0.133 – **threshold not met, continue resizing block**

New Sentiment (pos, neu, neg): (0.157, 0.399, 0.444)

Elastic-Block-3: I have had multiple conversations with your support. It has been a complete waste of time. No one has been able to address my problem.

Subjectivity =0.33 – **threshold MET, stop processing block**

New Sentiment (pos, neu, neg): (0.016, 0.188, 0.796)

Overall sentiment for the dynamically expanded block (“Elastic-Block-3”) is reported as negative given the high “0.796” value, which leads to more accurate sentiment insight into the actual conversation.

Note: the scope of this example is limited to a transcript that begins with text indicating a neutral sentiment followed by additional text that leans towards a negative sentiment. However, this solution is not limited to such an example. The solution can process transcripts where the additional text leans towards a positive sentiment, or where the initial text starts with a negative or positive sentiment.

them with the overall schematic figure above so that reviewers can follow along in the figure.

Tip: Solutions typically include conventional components and new components. Because the operation of conventional components is well-known, a brief description of the conventional components is enough. However, for new components, a thorough explanation is needed. In this case, STT engines, sentiment analysis engines, and subjectivity engine are well known, so their descriptions were relatively brief. The meta-learning model and the scoring module in the context of this solution are new components, so more detail was warranted.

G. Section Guide

Tip: in view of the example identified in Section 1.1 that clearly shows the technical problem, provide a corresponding example that clearly shows the technical solution. In this case, and by using the same example transcript, reviewers are informed about the expanding text block that solves the limitations of conventional tools that generally split the transcript into static text blocks and limit the analysis to each of these static text blocks.

1.3 How is the invention new or “novel” compared to known solutions? Consider whether the underlying technical methodology is unique compared to existing solutions, even those in different industries. Please identify which steps described in Section 1.2 are novel.

How the invention is new:

The technique of determining the regularized sentiment score, and using the score as a trigger in dynamically expanding a text block as described in steps 7 and 8 is new.

Even though sentiment analysis engines and the subjectivity engine are known components in isolation, simply combining their respective outputs would not overcome the technical problems identified in Section 1.1. First of all, simply combining their respective outputs would lead to multiple scores in accordance with the number of sentiment analysis engines. Second, the multiple scores would still not solve the technical problem related to large variance because there would still be a large variation in the multiple scores. Using a meta-learning model to find a representative sentiment probability across the sentiment analysis engines (i.e., p_m), combined with factoring in the subjectivity probability (i.e., C_{subj}) from the subjectivity engine is what precisely overcomes the technical problems. Moreover, even if somehow it was known to use these different types of engines and a meta-learning model to produce a single score, no one has contemplated using the score to dynamically expand a text block, because text blocks are typically static in nature.

Known art and how this invention differs:

1. Methods described in <insert hyperlink of known reference, e.g., website, published paper> adjust the final sentiment score as a function of sentiment scores of multiple text blocks, and thus do not dynamically resize a single text block by considering subjectivity.

2. Methods described in <insert hyperlink of known reference, e.g., website, published paper> separate text using the holder and topic methodology, and thus do not describe utilizing scoring of sentiment and subjectivity to arrive at a score that is then used to dynamically adjust a text block size.

H. Section Guide

Reviewers of this document will be looking for a combination of inventive steps with technical details supporting those steps.

Tip: In deciding what is “new,” consider whether your solution has at least one aspect that is different than a known solution. Also consider explaining why combining known solutions would still not result in your solution, or solve the same technical problems. Some ways to explain this include:

- *The combination of known solutions still does not result in our solution (see example on the left);*
- *One of the known solutions would have to be significantly re-engineered to work with the other solution(s).*

Tip: If the new aspect(s) were already highlighted in other sections, simply refer to the steps here.

Tip: the new aspect(s) of your invention should be fairly specific. Avoid saying that the combination of all of the components is what makes the solution new. In this case, although some conventional components were a part of the solution, the new aspect(s) can be pinpointed to steps 7-8.

1.4 Describe any alternative ways of operating or structuring the invention. Is there more than one way to practice any of the steps of your invention described in Section 1.2? If so, provide examples of those variations.

Rather than using a regularized sentiment scoring module, the subjectivity probability (C_{subj}) for the first text block output by the subjectivity engine can be input into the meta-learning model. The meta-learning model can be trained using conventional supervised classification techniques to account for the subjectivity probability (C_{subj}) when determining the sentiment probability (p_m). However, because training the meta-learning model to incorporate the subjectivity probability (C_{subj}) requires additional compute power and processing time, the solution disclosed above with the regularized sentiment scoring module is the preferred approach.

I. Section Guide

Reviewers of this document will be looking for any alternative ways of implementation.

Tip: Focus on the new aspect(s) with respect to any potential alternatives. Identifying alternatives would prevent any competitors from approaching the technical problem with a relatively simple workaround.